# ICT Skills

## The big idea

**ICT (Information and Communication Technology)** skills are the everyday computer skills every job now needs: understanding the parts of a computer, using the operating system, managing files, keeping the computer healthy and secure, and using office tools like spreadsheets and presentations.

## Computer system and software

A computer system has **hardware** (physical parts – CPU, keyboard, monitor, printer) and **software** (programs).

| Software type | Purpose | Examples |
|---|---|---|
| **System software** | Runs the hardware and other programs | Operating systems – Windows, Linux (Ubuntu), macOS, Android |
| **Utility software** | Maintains and protects the computer | Antivirus, disk cleanup, backup, compression tools |
| **Application software** | Does user tasks | LibreOffice Writer/Calc/Impress, browsers, media players |

## Operating system and user interface

The **operating system (OS)** is the link between the user, applications and hardware. It manages memory, files, processes and devices, and provides a **user interface**:

- **CUI/CLI (Character/Command Line)** – type commands (e.g. MS-DOS, Linux terminal).
- **GUI (Graphical)** – icons, windows, menus and a mouse (Windows, Ubuntu desktop).
- **Touch, gesture and voice-based** interfaces – smartphones, smart speakers.

## Files and folders

A **file** stores data (a document, picture, song); its **extension** shows the type: `.odt` (Writer), `.ods` (Calc), `.odp` (Impress), `.odb` (Base), `.jpg`, `.mp3`. Files are kept in **folders (directories)**.

COMMON FILE ACTIONS IN UBUNTU/WINDOWS

- New folder: right-click empty space → **New Folder**.
- Rename: select the file → **F2** → type the new name → Enter.
- Copy/Move: **Ctrl+C** / **Ctrl+X**, then **Ctrl+V** in the new folder.
- Delete: select → **Delete** key (it goes to Recycle Bin/Trash; **Shift+Delete** deletes permanently).

## Computer care and maintenance

- **Clean** keyboard, screen and vents regularly; keep food and liquids away.
- Run **Disk Cleanup** and **defragmentation** (for hard disks), delete unwanted files.
- Keep the OS and software **updated**.
- Take regular **backups** on an external drive or cloud.
- Scan with **antivirus**; do not open unknown attachments.

## Computer security and privacy

Threats you must know:

- **Virus** – attaches itself to files and spreads when they are opened.
- **Worm** – spreads by itself through networks.
- **Trojan horse** – looks useful but secretly harms the system.
- **Spyware** – secretly collects your data; **adware** shows unwanted ads.
- **Phishing** – fake e-mails/websites that trick you into giving passwords or bank details.

Protection: antivirus, **firewall**, strong passwords, updates, avoiding unknown links, logging out of public computers, using only secure (**https**) sites for payments.

**A WORM IS NOT A VIRUS**

A virus needs a host file and a user action to spread; a worm spreads on its own over the network.

## Spreadsheet and presentation basics

- **Spreadsheet (LibreOffice Calc)**: data in **cells** named by column letter + row number (B3). Formulas start with `=`, e.g. `=SUM(B2:B10)`, `=AVERAGE(C2:C6)`. Insert a chart: select data → **Insert > Chart**.
- **Presentation (LibreOffice Impress)**: new slide **Slide > New Slide** (Ctrl+M); choose a layout from the **Layouts** panel of the sidebar; add transitions in the **Slide Transition** panel; start the show with **F5**.

**KEYBOARD SHORTCUTS SCORE EASY MARKS**

Ctrl+S save, Ctrl+Z undo, Ctrl+Y redo, Ctrl+A select all, Ctrl+P print, F5 start slide show.

## Quick recap

- Software: system, utility, application. OS manages hardware and gives a user interface.
- Interfaces: CUI, GUI, touch, voice, gesture.
- Maintain by cleaning, updating, backing up and scanning.
- Threats: virus, worm, trojan, spyware, phishing; protect with antivirus, firewall, strong passwords.

## Practice questions

1. Which of the following is system software?
   (a) LibreOffice Writer (b) Ubuntu Linux (c) A web browser (d) A media player
2. An interface where the user types commands is called
   (a) CUI (Character User Interface) (b) GUI (c) Touch interface (d) Gesture interface
3. Malware that spreads through a network on its own, without a host file, is a
   (a) virus (b) trojan horse (c) worm (d) cookie
4. Which key combination permanently deletes a file without sending it to the Recycle Bin?
   (a) Ctrl+Delete (b) Alt+Delete (c) Ctrl+D (d) Shift+Delete
5. A fake bank e-mail asking you to "verify your password" is an example of
   (a) spamming (b) phishing (c) backup (d) defragmentation
6. The file extension of a LibreOffice Calc spreadsheet is
   (a) .odt (b) .odp (c) .ods (d) .odb
7. Differentiate between system software and application software with one example each.
8. List any three threats to computer security and three ways to protect a computer from them.

Answers: 1. b 2. a 3. c 4. d 5. b 6. c 7. see notes 8. see notes